

➤ **What is Android?**

1. Android is an open-source mobile operating system developed by Google. It is mainly used for smartphones and tablets, and it also powers devices such as smart TVs, smartwatches, and cars.
2. Android is software that manages a mobile device and allows users to run different applications.

➤ **Important Points About Android:-**

1. Operating System
 - Android is an Operating System (OS).
 - It works as a bridge between the user, applications, and hardware.
2. Developed by Google
 - Android is primarily developed and maintained by Google.
3. Open Source:
 - Android is largely based on the open-source Android Open Source Project (AOSP).
 - Developers and device manufacturers can customize Android for different devices.
4. Used on Different Devices:
 - Android is used in:
 - Smartphones
 - Tablets
 - Smart TVs
5. Provides User Interface
 - Android provides elements that users interact with, such as:
 - Buttons
 - Text fields
 - Menus
 - Notifications
 - Settings
 - Home screen
6. Runs Applications
 - Android provides an environment where applications can run.
 - Examples:
 - WhatsApp
 - YouTube
 - Instagram
 - Calculator

➤ **What is Android Development?**

1. Android Development is the process of designing, developing, testing, and maintaining applications that run on Android devices.
2. Android Development means creating mobile applications for Android devices such as smartphones, tablets, smart TVs, and smartwatches.

➤ **Key Points:**

1. Application Development
 - Android Development is used to create mobile applications.
 - Example: Calculator, Notes, Banking, Shopping, and Education apps.
2. Designing the App
 - Developers create the User Interface (UI) of the application.
 - Example:
 - Buttons
 - Text fields
 - Images
 - Menus
 - Lists
 - Navigation
3. Writing Code
 - Developers write code to define how the application works.
 - Common languages include Java and Kotlin.
4. Using Android Studio
 - Android Studio is the official IDE used for Android application development.
 - It provides tools for:
 - Writing code
 - Designing UI
 - Running applications
 - Debugging
 - Testing
5. Working with Device Features
 - Android applications can use device features such as:
 - Camera
 - GPS
 - Microphone
 - Speaker
 - Bluetooth
 - Internet

- Notifications

6. Testing the Application

- Developers test applications to find and fix errors.
- Testing can be done using:
 - Android Emulator
 - Real Android Device

7. Publishing the Application

- After development and testing, an application can be published on the Google Play Store

➤ **Examples:-**

Example 1: Calculator App

- Number buttons
- Addition, subtraction, multiplication, division
- Result display

Example 2: Notes App

- Add note
- Edit note
- Delete note
- Save notes
- Search notes

➤ **Android Developer:**

An Android Developer is a software developer who designs, develops, tests, and maintains applications for Android devices.

An Android Developer is a person who creates Android applications using programming languages and development tools such as Android Studio.

➤ **Android Developer Role:**

1. Understand the Requirement

- Understand what the application needs to do.
- Example: A college needs an app to manage student attendance.

2. Design the Application

- Create the user interface (UI).
- Example:
 - Login screen
 - Home screen

3. Write Code

- Write the logic that makes the application work.
- Common languages:
 - Java
 - Kotlin

4. Use Android Studio

- Android Studio is used to develop and manage Android projects.
- The android studio specifically used to develop the android application which executes on android OS.

5. Implement Features

- Add required functionality such as:
 - Login/Register
 - Database
 - Notifications
 - Camera

6. Test the Application

- Check whether the application works correctly.
- Testing can be performed on:
 - Real Android devices (Smartphones, tabs)
 - Android Emulator

7. Find and Fix Errors

- Identify bugs and solve them.
- Example: App crashes when the user clicks the Login button → developer finds and fixes the problem.

8. Maintain and Update the App

- Fix issues.
- Improve performance.
- Add new features.
- Update the application when required.

➤ Android Project Directory Structure:

Manifests -Contains AndroidManifest.xml, which provides important information about the application to Android. It also defines app components and required permissions.

Java – Contains the Java source code of the application. It is used to implement the app's logic and functionality.

Res (Resources) – Contains non-code resources used by the application. It includes folders such as layout, drawable, mipmap, and values.

Layout – Contains XML files that define the user interface of the application. It specifies the arrangement of buttons, text, images, and other UI elements.

Drawable – Contains images, vector graphics, and drawable XML resources. These resources are used to display or design graphical elements in the application.

Mipmap – Mainly contains the application's launcher icons. These icons are displayed on the device's home screen and app launcher.

Values – Contains reusable resources such as strings, colors, dimensions, and themes. It helps keep common values organized and reusable.

Gradle – Gradle is the build system used to build the Android application. It manages dependencies, compilation, and different build configurations.

build.gradle – Contains configuration and instructions used by Gradle to build the project or module. It can define plugins, dependencies, SDK settings, and other build options.

➤ **XML(Extensible Markup Language):**

XML (Extensible Markup Language) is used in Android to define and design the user interface (UI) of an application. It uses tags and attributes to describe UI elements such as TextView, Button, ImageView, and EditText.

The XML is used to design the UI of the application.

➤ **XML Elements & Main Attributes**

XML Element	Purpose	Main Attributes
TextView	Displays text on the screen	android:id, android:text, android:textSize, android:textColor, android:gravity
EditText	Allows the user to enter text	android:id, android:hint, android:inputType, android:text, android:textSize
Button	Creates a clickable button	android:id, android:text, android:onClick, android:enabled
ImageView	Displays an image	android:id, android:src, android:scaleType, android:contentDescription
ImageButton	Creates a clickable image button	android:id, android:src, android:contentDescription, android:background
CheckBox	Allows selecting multiple options	android:id, android:text,

XML Element	Purpose	Main Attributes
		android:checked, android:enabled
RadioButton	Allows selecting one option from a group	android:id, android:text, android:checked
RadioGroup	Groups RadioButtons together	android:id, android:orientation
Switch	Provides an ON/OFF option	android:id, android:text, android:checked, android:enabled
ToggleButton	Provides ON/OFF toggle functionality	android:id, android:textOn, android:textOff, android:checked
ProgressBar	Shows progress/loading	android:id, android:progress, android:max, android:indeterminate
SeekBar	Allows selecting a value by sliding	android:id, android:progress, android:max
Spinner	Provides a dropdown list	android:id, android:entries, android:spinnerMode
AutoCompleteTextView	Provides text suggestions while typing	android:id, android:hint, android:inputType
ScrollView	Makes content	android:id, android:fillViewport

XML Element	Purpose	Main Attributes
	vertically scrollable	
HorizontalScrollView	Makes content horizontally scrollable	android:id, android:fillViewport
WebView	Displays web content inside the app	android:id, android:layout_width, android:layout_height